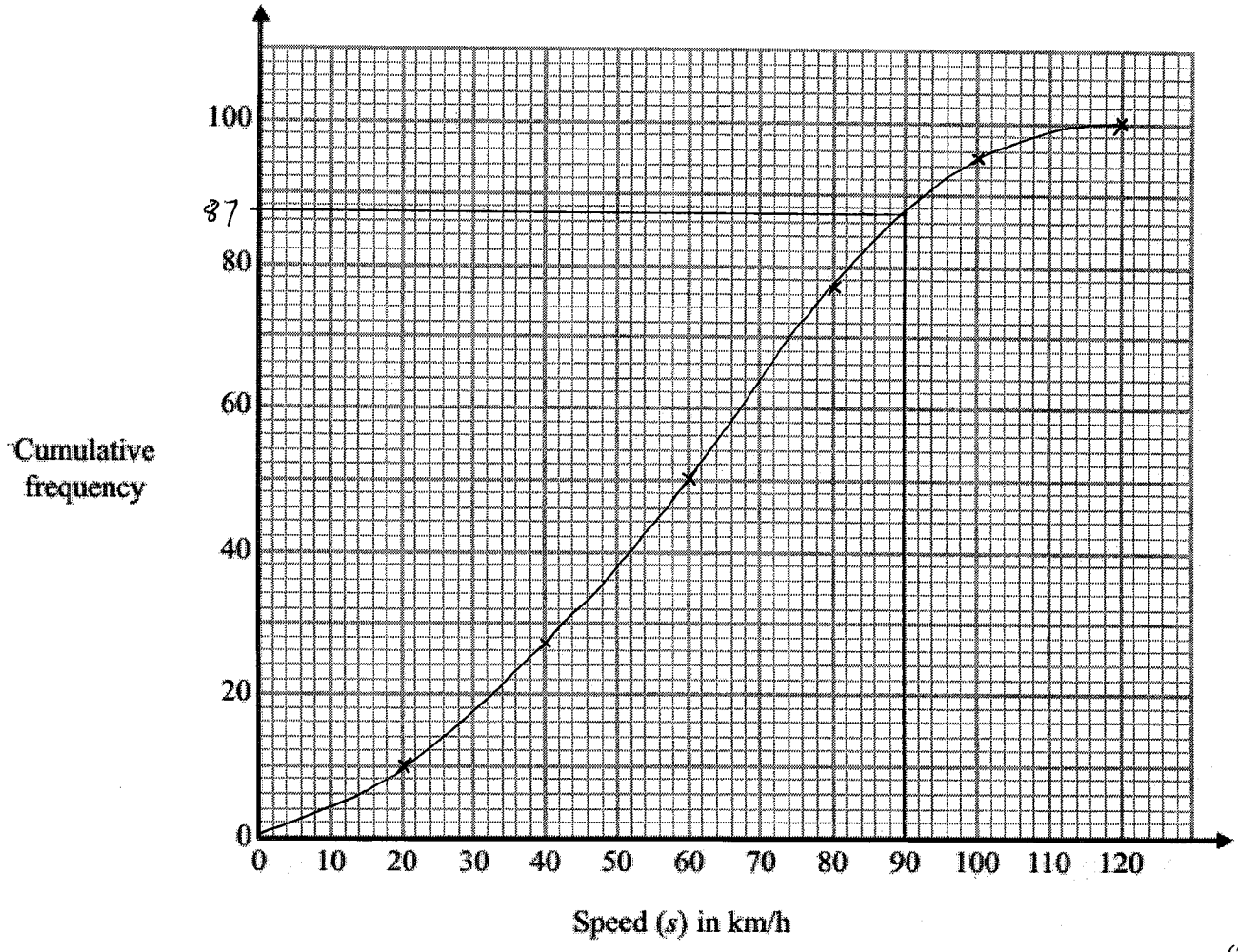
1 The frequency table shows the speeds of 100 cars.

Speed (km/h)	Frequency	C.F
$0 < s \leqslant 20$	10	10
$20 < s \leqslant 40$	17	27
$40 < s \leqslant 60$	23	50
$60 < s \leqslant 80$	27	77
$80 < s \leqslant 100$	18	95
$100 < s \leqslant 120$	5	100

(a) On the grid, plot a cumulative frequency graph for this information.



(b) Find an estimate for the number of cars travelling over 90 km/h. (2)

100 - 87 = 13

13

(12 – 14) (1)

(Total for Question 1 is 3 marks)

2 Expand and Simplify $(x + 2)(2x + 3)(3x + 1)$

$(2x^2 + 3x + 4x + 6)(3x + 1)$

$(2x^2 + 7x + 6)(3x + 1)$

$6x^3 + 2x^2 + 21x^2 + 7x + 18x + 6$

$6x^3 + 23x^2 + 25x + 6$

$6x^3 + 23x^2 + 25x + 6$

(Total for Question 2 is 3 marks)

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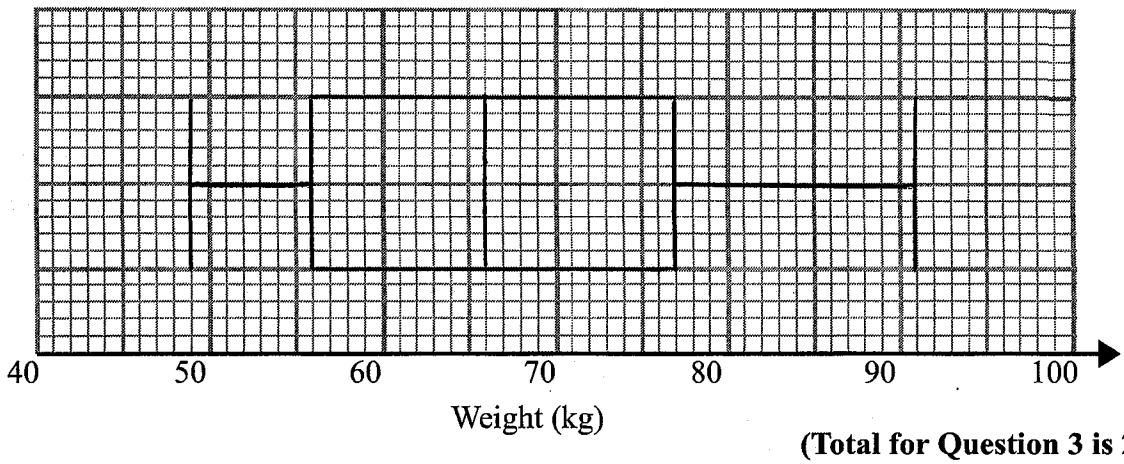
3 The weights of 11 pigs, in kg, are recorded below.

49 52 56 62 66 66 69 71 77 78 91

Draw a box plot for this information.



(Total for Question 3 is 2 marks)

4 Write $5\sqrt{75}$ in the form $k\sqrt{3}$, where k is an integer.

$5 \times \sqrt{25} \times \sqrt{3}$

$5 \times 5 \times \sqrt{3}$

$25 \times \sqrt{3}$

$25\sqrt{3}$

$25\sqrt{3}$

(Total for Question 4 is 2 marks)

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5 Find the value of $\left(\frac{27}{125}\right)^{-\frac{2}{3}}$

$\left(\frac{3}{5}\right)^{-2} = \left(\frac{9}{25}\right)^{-1}$

$\frac{25}{9}$

(Total for Question 5 is 2 marks)

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6 Make x the subject of the formula $ax + y = 4x - 2b$

$ax + y + 2b = 4x$

$y + 2b = 4x - ax$

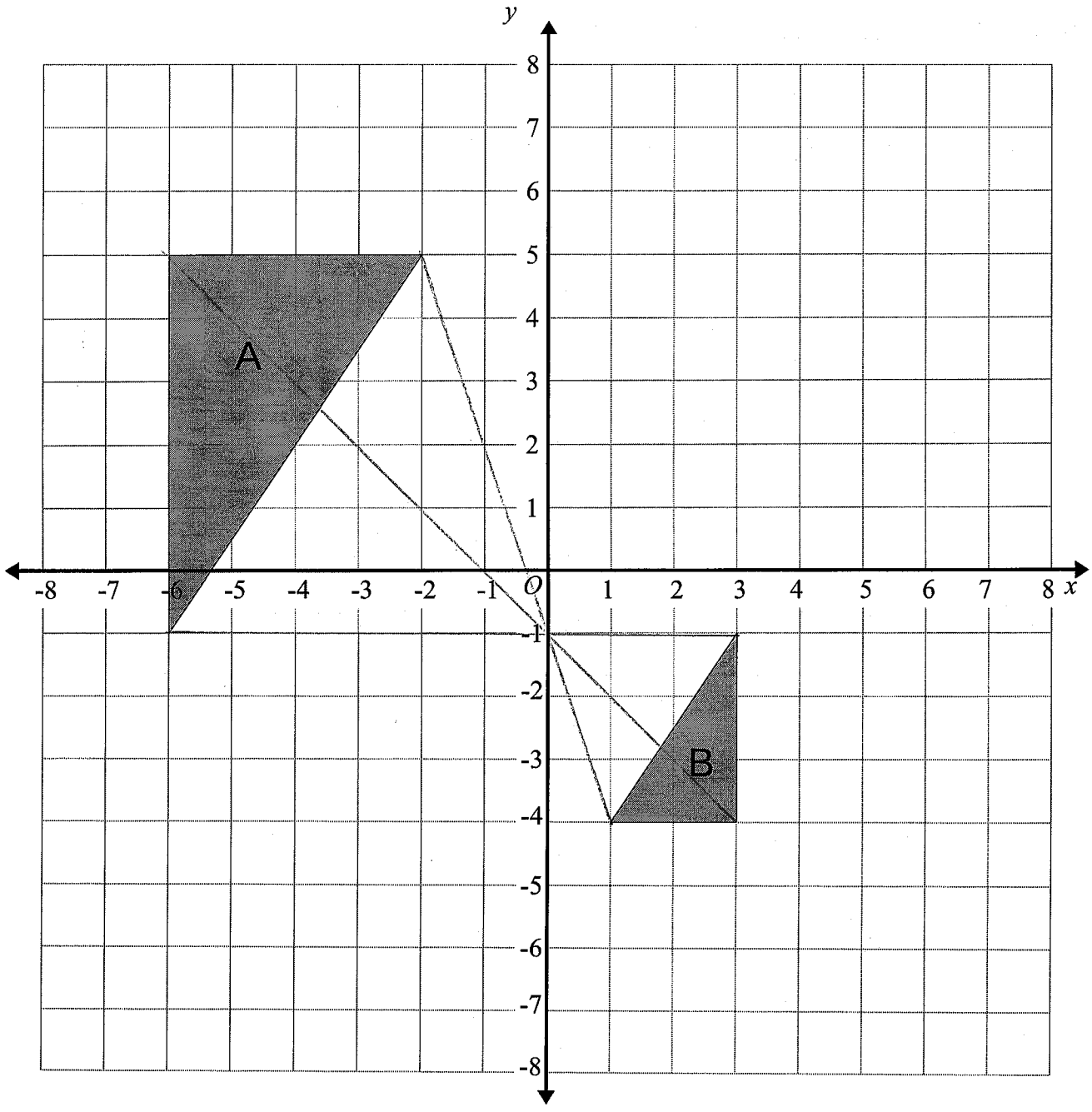
$y + 2b = x(4 - a)$

$\frac{y + 2b}{4 - a} = x$

$x = \frac{y + 2b}{4 - a}$

(Total for Question 6 is 3 marks)

7



Describe fully the single transformation which maps triangle A onto triangle B.

Enlargement, scale factor $-\frac{1}{2}$,
centre $(0, -1)$

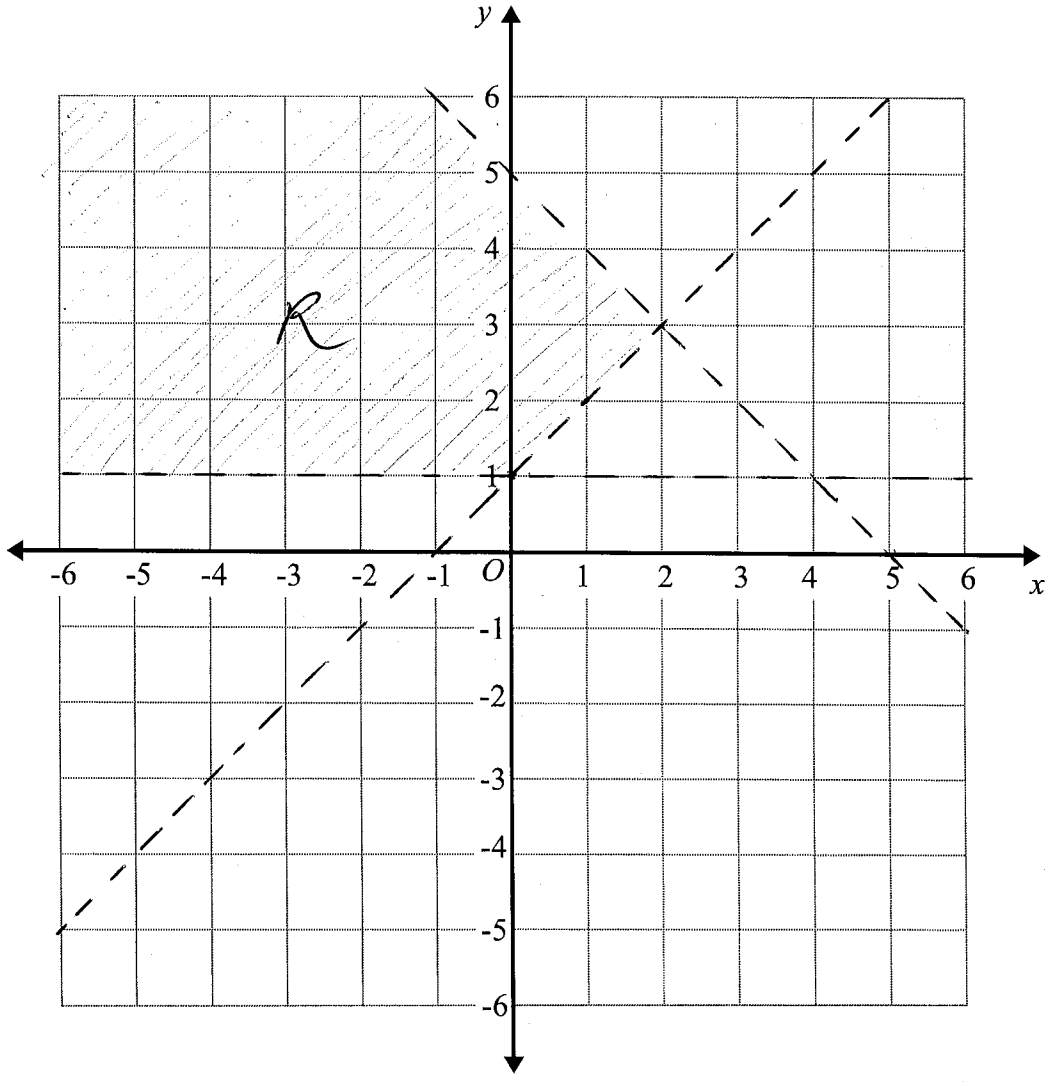
(Total for Question 7 is 2 marks)

8 On the grid shade the region that satisfies all these inequalities

$x + y < 5$ $\qquad$ $y > x + 1$ $\qquad$ $y > 1$

Label the region **R**. (5, 0)
(0, 5)

x	0	1	2
y	1	2	3



(Total for Question 8 is 3 marks)

9 Write $0.\dot{4}\dot{5}$ as a fraction in its simplest form.

$0.\dot{4}\dot{5} = x$

$45.\dot{4}\dot{5} = 100x$

$45 = 99x$

$x = \frac{45}{99}$

$x = \frac{45}{99} = \frac{5}{11}$

$\frac{5}{11}$

(Total for Question 9 is 2 marks)

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10 Line A passes through the points (2, 1) and (8, 10)
Line B passes through the points (1, 7) and (-3, 1)
Show that Line A and Line B are parallel.

$x_1 \; y_1 \quad x_2 \; y_2$

Line A

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{10 - 1}{8 - 2}$

$= \frac{9}{6}$

$= \frac{3}{2}$

Line B

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{1 - 7}{-3 - 1}$

$= \frac{-6}{-4}$

$= \frac{3}{2}$

Same gradient $\therefore$ parallel

(Total for Question 10 is 3 marks)